

Review on Probability

Random variable (r.v.)

The r.v. X takes a value in $\mathcal{X}$ (set of possible values) according to the outcome of an event.

E.g. Event: flip a coin

$$X = \{H, T\}$$

X has value H when heads appears, T when tails appears.

Associated with r.v. X is a distribution $P(X=x)$ describes the frequency of the events occurring.

Examples:

Discrete

indicator variable # of people in room

$$\gamma = \{0, 1\}$$

$$X = \mathbb{Z}_+ \text{ (non-negative integers)}$$

pmf = Probability Mass Function

$P(X=x)$ = probability of taking x

$$\sum_{x \in X} p(X=x) = 1$$

$$0 \leq P(X=x) \leq 1 \quad \forall x \in \mathcal{X}$$

Continuous

sensor ready $\mathcal{X} = \mathbb{R}$

pdf = probability density function

$p(x)$ = likelihood of x

$$P(a \leq x \leq b) = \int_a^b p(x) dx$$

$$\int_{\mathcal{X}} p(x) dx = 1$$

$$0 \leq p(x) \quad \forall x \in X$$

Note: $P(X=x) = p(x)$ or $p_x(x)$
↑
r.v.

Example Distributions

- Bernoulli (com)

$\mathcal{X} = \{0, 1\}$, $\pi = \text{probability that } 1 \text{ occurs } (0 \leq \pi \leq 1)$

$$P(X=1) = \pi$$

$$P(X=0) = 1 - \pi$$

$$p(x) = \pi^x (1-\pi)^{1-x}$$

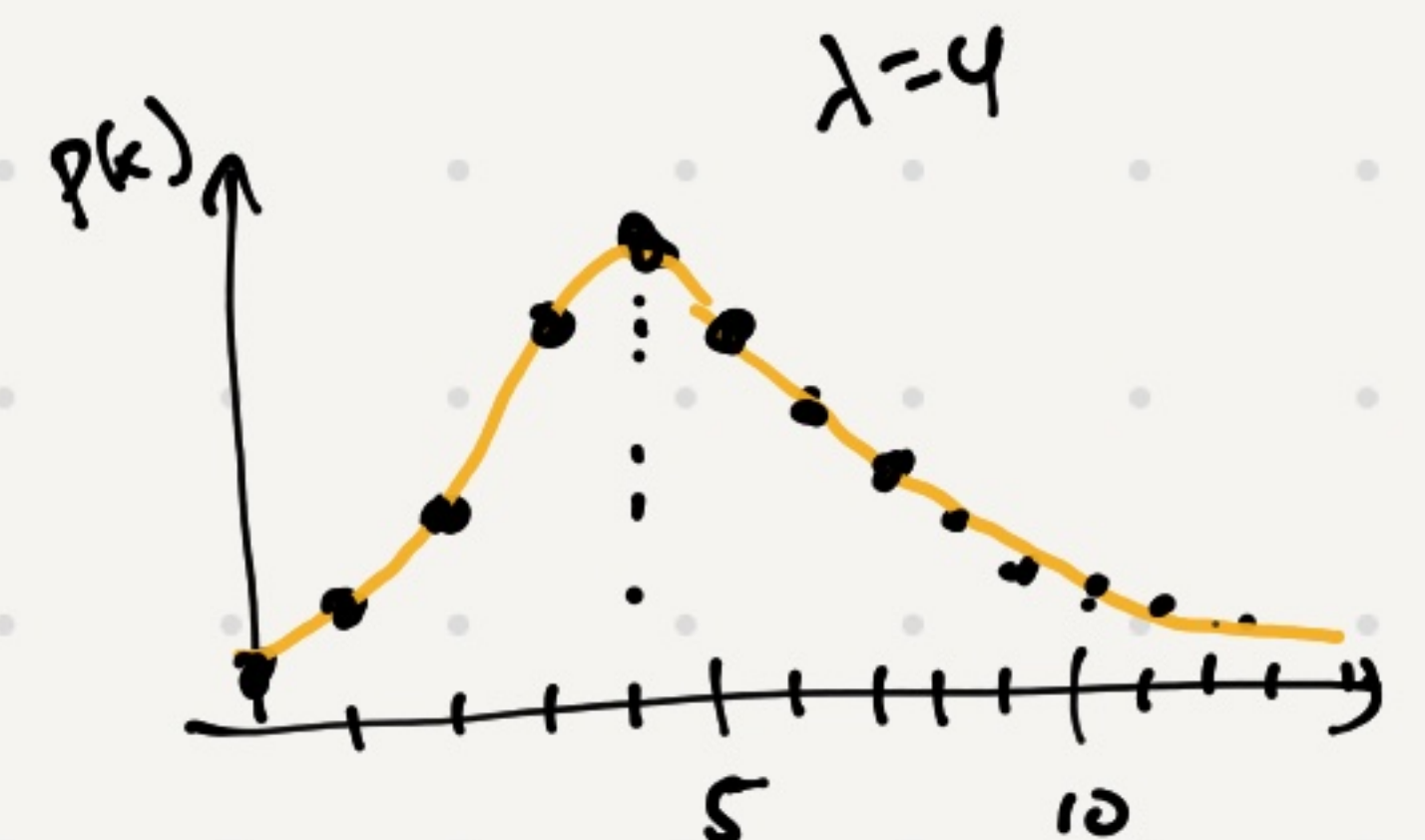
$$\begin{aligned} x=0: p(0) &= \pi^0 (1-\pi)^{1-0} = 1-\pi \\ x=1: p(1) &= \pi^1 (1-\pi)^{0} = \pi \end{aligned}$$

- Poisson : # of arrivals over a fixed time.

$$X = \mathbb{Z}_+, \quad \lambda = \text{average arrival rate} \geq 0$$

$$p(x) = \frac{1}{x!} e^{-\lambda} \lambda^x$$

$$x! = x \cdot (x-1) \cdot \dots \cdot 1$$



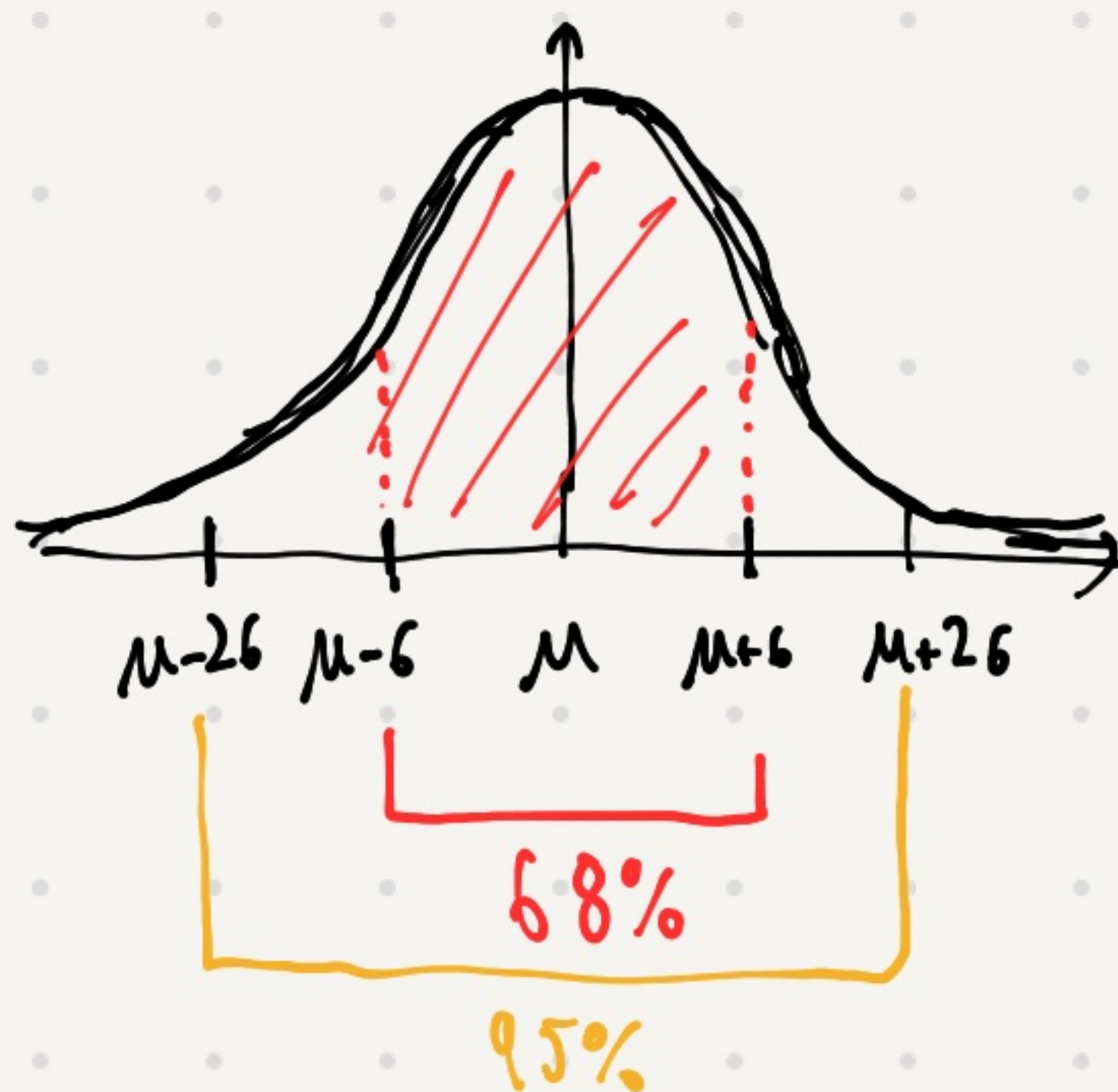
Gaussian (Normal)

$X \in \mathbb{R}$

μ = mean

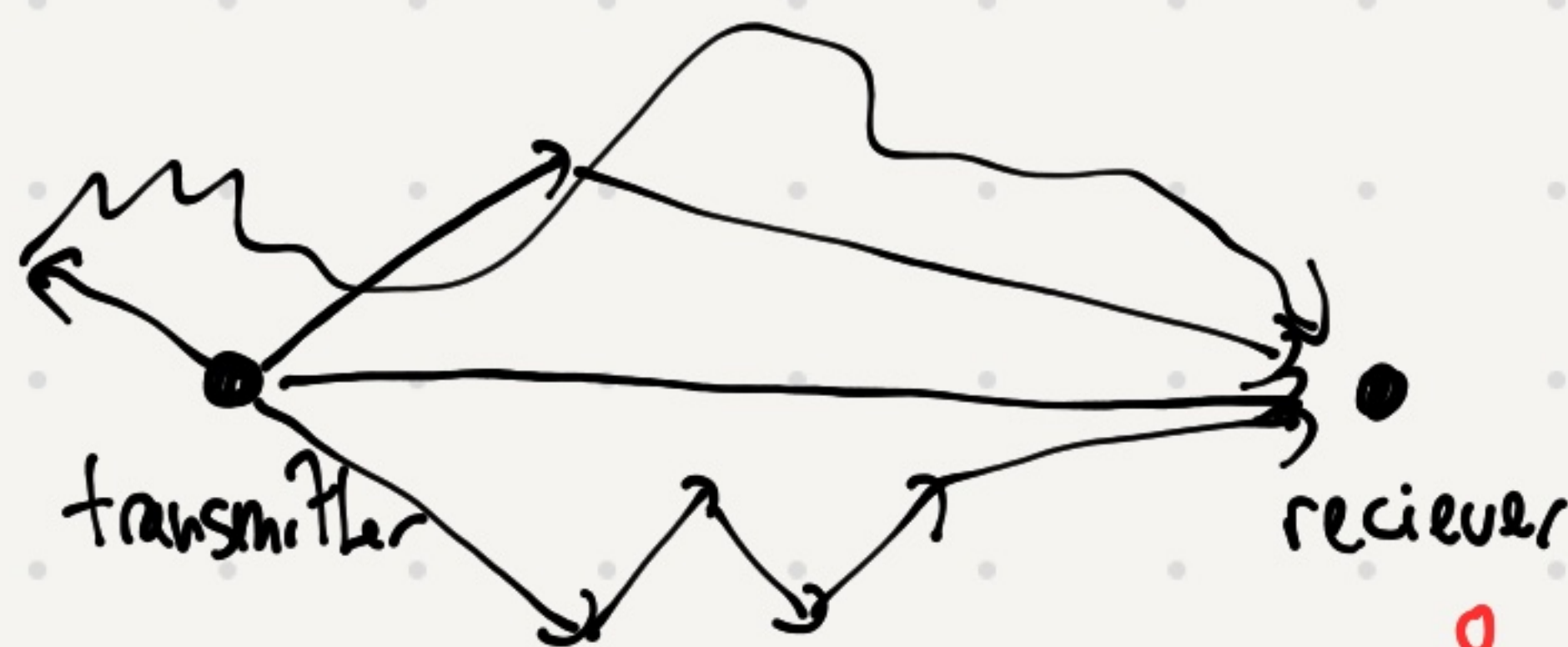
σ^2 = variance $> 0 \Rightarrow \sigma$ = Standard deviation

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} = N(x|\mu, \sigma^2)$$

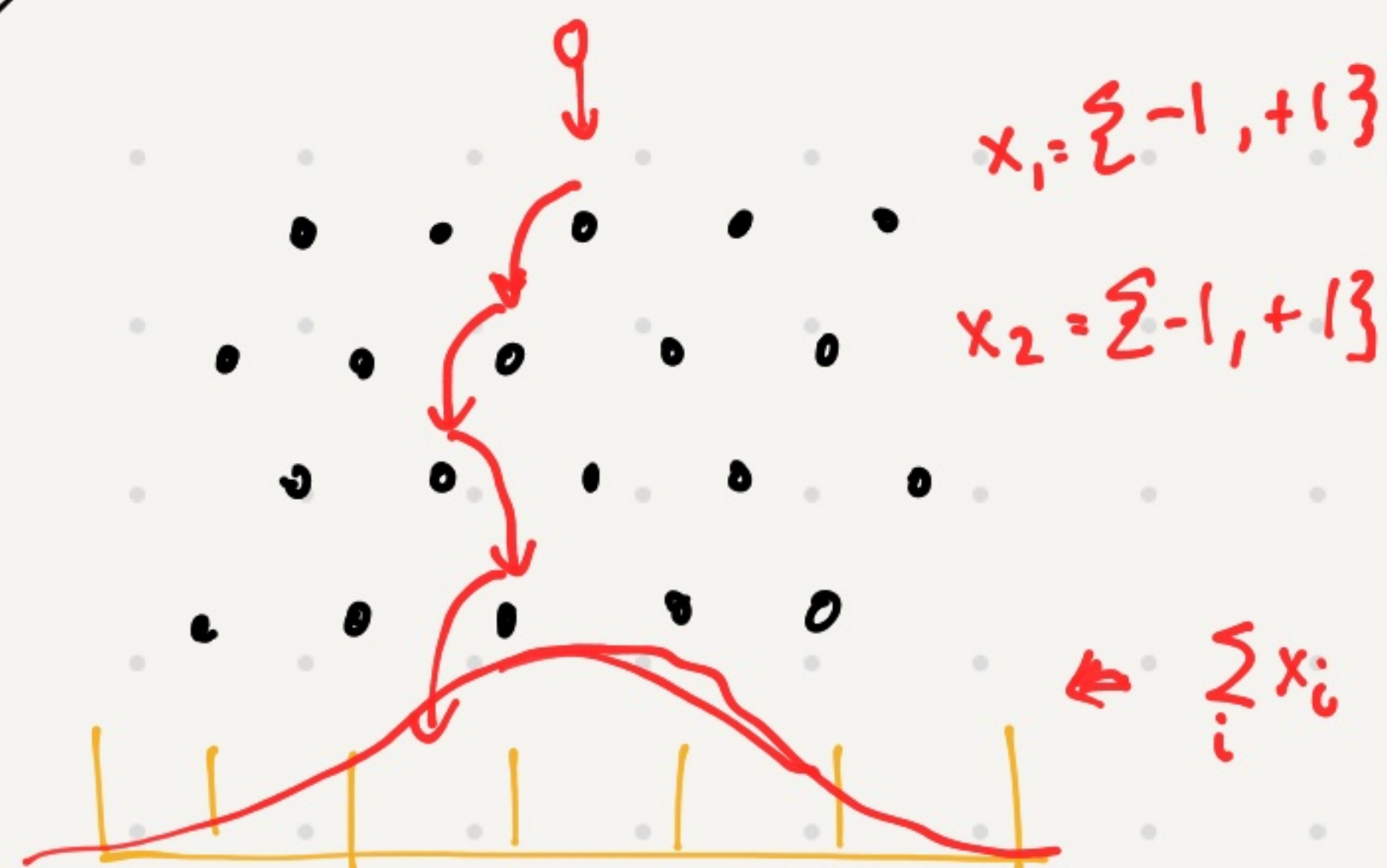


Central Limit Theorem (CLT)

Sum of N r.v. $\rightarrow$ Gaussian distribution for large N



Science museum:



Joint distribution - distribution of more than 2 r.v.

Probability/likelihood of $X=x$ and $Y=y$

$$P(X=x, Y=y) = p(x, y)$$

Marginal distribution

dist of one variable in the joint.

$$P(X=x) = \sum_{y \in \mathcal{Y}} p(X=x, Y=y)$$

$$p(x) = \int_{y \in \mathcal{Y}} p(x, y) dy$$

"Sum out the other r.v."
"marginalization"

Example: $X \in \{0, 1\}$, $Y \in \{0, 1\}$

$p(x, y)$	$Y=0$	$Y=1$	$P(X)$
$X=0$	0.08	0.12	0.2
$X=1$	0.32	0.48	0.8
$p(y)$	0.4	0.6	

"margin of table"

Conditional Distribution

distribution of one r.v. when the value of another r.v. is known (given).

$$P(X=x | Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$$

condition
"given"

Eg. $P(X=0|Y=0) = \frac{0.08}{0.4} = 0.2$
 $P(X=1|Y=0) = \frac{0.32}{0.4} = 0.8$
 sum to 1

$$\Rightarrow p(x, y) = p(x|y)p(y)$$

Statistical Independence

dist of one r.v. does not change when knowing the value of another r.v.

$$X \perp Y \text{ i.f.f. } p(x|y) = p(x)$$

$$X \perp Y \text{ i.f.f. } p(x, y) = p(x)p(y)$$

(joint is the product of marginals)

Eg. in our example
 $X \perp Y$.

Bayes' Rule

$$p(x, y) = p(x|y)p(y)$$

$$p(x, y) = p(y|x)p(x)$$

$$p(y|x)p(x) = p(x|y)p(y)$$

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

$$p(x) = \int_y p(x, y) dy = \int_y p(x|y)p(y) dy$$

$$\Rightarrow p(y|x) = \frac{p(x|y)p(y)}{\int p(x|y)p(y) dy}$$

Bayes' Rule

"Given only $p(x|y)$ & $p(y)$, we can 'invert' the conditioning to get $p(y|x)$."

Expectations

Suppose we have a function $f(x)$ & r.v. X .

On average, what value of $f(X)$ would I observe?

$$E_x[f(x)] = \sum_{x \in \mathcal{X}} f(x) p(x) \quad \begin{array}{c} \uparrow \quad \uparrow \\ \text{weight} \end{array} \quad \text{"weighted average of function values"}$$

$$E_x[f(x)] = \int_{x \in \mathcal{X}} f(x) p(x) dx$$

• mean: $E[x] = \int x p(x) dx = \mu_x$

• variance: $\text{Var}(x) = E[(x - \underbrace{E[x]}_{\mu_x})^2] = E[x^2] - (E[x])^2 = \sigma_x^2$

• covariance: $\text{Cov}(x, y) = E_{x, y}[(x - E_x)(y - E_y)]$
 $= \iint (x - E_x)(y - E_y) dx dy = E[xy] - (E_x)(E_y)$
 $= \sigma_{xy}^2$

• conditional expectation:

$$E_{x|y}[x] = \int x \underbrace{p(x|y)}_{\text{conditional dist.}} dx$$

Linear Algebra Review

column vector: $x \in \mathbb{R}^d$ $x = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$ usually we are column-centric.

matrix: $A \in \mathbb{R}^{m \times n}$ $A = \begin{bmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{bmatrix} = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$
 $\uparrow$ a_i is the i th column of A
 $a_i \in \mathbb{R}^m$

inner product: $x^T y = \sum_{i=1}^d x_i y_i$ (similarity b/w x & y)

length (norm): $\|x\| = \sqrt{x^T x} = \sqrt{\sum_{i=1}^d x_i^2}$

distance (metric): $\|x - y\| = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$

outer product: $x y^T = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} \begin{bmatrix} y_1 & \dots & y_d \end{bmatrix} = \begin{bmatrix} x_1 y_1 & \dots & x_1 y_d \\ \vdots & \ddots & \vdots \\ x_d y_1 & \dots & x_d y_d \end{bmatrix}$
 "matrix of pairwise products"

Matrix-vector multiply

① $A \in \mathbb{R}^{m \times d}$
 $x \in \mathbb{R}^d$: $y = Ax = \begin{bmatrix} | & \dots & | \\ a_1 & \dots & a_d \\ | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = \sum_{i=1}^d x_i a_i \in \mathbb{R}^m$
 linear combo of columns of A with weights in x .

② $A \in \mathbb{R}^{d \times m}$
 $x \in \mathbb{R}^d$: $y = A^T x = \begin{bmatrix} | & \dots & | \\ a_1 & \dots & a_m \\ | & & | \end{bmatrix}^T x = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix} x$
 $= \begin{bmatrix} a_1^T x \\ \vdots \\ a_m^T x \end{bmatrix} \in \mathbb{R}^m$ vector of inner products b/w columns of A & x .

Matrix-matrix multiply

① $A \in \mathbb{R}^{m \times d}$
 $B \in \mathbb{R}^{d \times n}$: $AB = A \begin{bmatrix} | & \dots & | \\ b_1 & \dots & b_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & \dots & | \\ Ab_1 & \dots & Ab_n \\ | & & | \end{bmatrix} \in \mathbb{R}^{m \times n}$

A multiplied w/ each column of b

② $A \in \mathbb{R}^{d \times m}$
 $B \in \mathbb{R}^{d \times n}$: $A^T B = \begin{bmatrix} -a_1^T- \\ \vdots \\ -a_m^T- \end{bmatrix} \begin{bmatrix} | & \dots & | \\ b_1 & \dots & b_n \\ | & & | \end{bmatrix} = \begin{bmatrix} a_i^T b_j \end{bmatrix}_{ij} \in \mathbb{R}^{m \times n}$

matrix of column-wise inner products.

③ $A \in \mathbb{R}^{m \times d}$
 $B \in \mathbb{R}^{h \times d}$: $AB^T = \begin{bmatrix} | & \dots & | \\ a_1 & \dots & a_d \\ | & & | \end{bmatrix} \begin{bmatrix} -b_1^T- \\ \vdots \\ -b_d^T- \end{bmatrix} = \sum_{i=1}^d a_i b_i^T \in \mathbb{R}^{m \times h}$

Sum of outerproducts of columns of A & B .

Vector r.v.

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} \quad \mathcal{X} = \mathbb{R}^d$$

Notation

$$p(x_1, x_2, \dots, x_d) = p(x)$$

$$\int p(x) dx = 1 = \int_{x_1} \int_{x_2} \dots \int_{x_d} p(x) dx_1 \dots dx_d = 1$$

mean vector: $\mu = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_d \end{bmatrix} = E[X] = \int_{\mathbb{R}^d} x p(x) dx$

Covariance matrix: $\text{cov}(x) = E[(X - E_X)(X - E_X)^T]$
 $= E[XX^T] - (E_X)(E_X)^T$

2-d example:

$$\text{cov}(x) = E \begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) \\ (x_2 - \mu_2)(x_1 - \mu_1) & (x_2 - \mu_2)^2 \end{bmatrix}$$
$$= \begin{bmatrix} \sigma_1^2 & \sigma_{12}^2 \\ \sigma_{12}^2 & \sigma_2^2 \end{bmatrix}$$

Annotations: σ_{12}^2 (off-diagonal) are covariances. σ_1^2 and σ_2^2 (diagonal) are variances.

Multivariate Gaussian

$$X \in \mathbb{R}^d, \text{ mean vector } \mu \in \mathbb{R}^d$$

$$\text{covariance matrix } \Sigma \in \mathbb{R}_{++}^d \quad (d\text{-dim positive definite symmetric matrix})$$

$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} \|x - \mu\|_{\Sigma}^2} = N(x | \mu, \Sigma)$$

determinant: $|\Sigma|$ = "volume of Gaussian"

Mahalanobis
Square-distance: $\|x - \mu\|_{\Sigma}^2 = (x - \mu)^T \Sigma^{-1} (x - \mu)$

Special Case Diagonal $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_d^2 \end{bmatrix}$ $\leftarrow \sigma_{ij}^2 = 0 \quad \forall i \neq j$

$$\Rightarrow N(x | \mu, \Sigma) = \prod_{i=1}^d N(x_i | \mu_i, \sigma_i^2)$$

Annotations: joint dist of x = product of marginal of x_i $\Rightarrow x_i$'s are statistically independent.

